

PROJECT REPORT ON

A Compact Dual-Band Dual-Polarized

Microstrip Antenna Array for MIMO-SAR

Applications

Submitted in partial fulfillment of the requirements for the award of

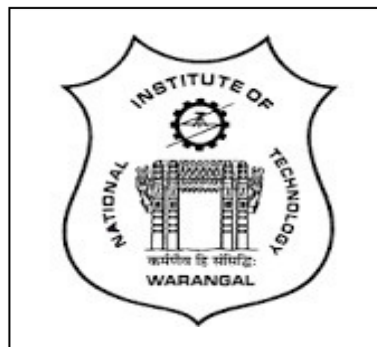
the degree of

BACHELOR OF TECHNOLOGY

IN

ELECTRONICS AND COMMUNICATION

ENGINEERING



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

NATIONAL INSTITUTE OF TECHNOLOGY WARANGAL

Submitted By:

Deepika Dwivedi

Roll No.: 24116028

Guided By:

Prof. Amarjit Kumar

Department of ECE

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my respected mentor Dr Amarjit Kumar, for providing me with the opportunity to work on this project and for their continuous guidance, valuable suggestions, and encouragement throughout the internship period. I am also thankful to National Institute of Technology, Warangal for offering me a platform to gain practical knowledge and hands-on experience in the field of antenna design and simulation. I extend my heartfelt thanks to my faculty members for their constant support and motivation, which helped me in successfully completing this internship. Lastly, I would like to thank my family and friends for their encouragement and support during this period.

Deepika Dwivedi

ABSTRACT

This report presents the design and implementation of a compact dual-band dual-polarized microstrip antenna array intended for MIMO-SAR (Multiple Input Multiple Output – Synthetic Aperture Radar) applications.

The primary objective of this work is to develop an antenna system capable of operating at two distinct frequency bands while maintaining high performance in terms of gain, bandwidth, and isolation. Dual-band functionality allows the antenna to support multiple radar frequencies, thereby improving imaging capabilities and system flexibility.

In addition, dual polarization is incorporated in the antenna design to enhance signal diversity and reduce interference between channels, which is a critical requirement in MIMO systems. The antenna array is designed using a corporate feeding network, ensuring uniform power distribution among all radiating elements.

The antenna structure is implemented using microstrip technology, which provides advantages such as compact size, low cost, ease of fabrication, and compatibility with printed circuit boards. The design and simulation are carried out using CST Microwave Studio, where key parameters such as return loss (S-parameters), gain, radiation pattern, and bandwidth are analyzed.

The simulated results demonstrate that the proposed antenna achieves good impedance matching at both frequency bands, stable radiation characteristics, and effective isolation between ports. These features make it suitable for applications in radar imaging, remote sensing, and satellite communication systems.

TABLE OF CONTENTS

1. Introduction	7
2. Literature Review	9
3. Objective of the Work	11
4. Methodology	12
4.1 Antenna Design Overview	
4.2 Substrate Selection	
4.3 Patch Design	
4.4 Feeding Network Design	
4.5 Dual-Band Operation	
4.6 Dual Polarization Technique	
5. Simulation Setup (CST)	18
6. Results and Analysis	20
6.1 Return Loss (S-Parameters)	
6.2 Gain and Radiation Pattern	
6.3 Bandwidth Analysis	
6.4 Isolation Performance	
7. Flowchart of Implementation	24
8. Advantages and Applications	25
9. Challenges Faced	26
10. Conclusion	27
11. Future Scope	28
12. References	29

INTRODUCTION

With the rapid advancement in wireless communication and radar technologies, there is a growing demand for antenna systems that are compact, efficient, and capable of supporting multiple functionalities. One such critical application area is Synthetic Aperture Radar (SAR), which is widely used for high-resolution imaging in remote sensing, surveillance, and environmental monitoring.

In recent years, the integration of Multiple Input Multiple Output (MIMO) technology with SAR systems has significantly enhanced system performance. MIMO technology enables the use of multiple transmitting and receiving antennas, which improves spatial resolution, signal quality, and data reliability. However, implementing MIMO-SAR systems requires antennas that can operate efficiently with multiple channels while maintaining low mutual coupling and high isolation.

Microstrip antennas have become a preferred choice for such applications due to their inherent advantages, including:

- Low profile and compact structure
- Lightweight and conformal design
- Ease of fabrication using PCB technology
- Cost-effectiveness

Despite these advantages, conventional microstrip antennas suffer from certain limitations such as:

- Narrow bandwidth
- Low gain
- Limited efficiency

To overcome these challenges, advanced antenna design techniques are required.

Proposed Approach

This work focuses on the design and implementation of a compact dual-band dual-polarized microstrip antenna array, specifically developed for MIMO-SAR applications.

The key features of the proposed antenna system include:

1. Dual-Band Operation

The antenna is designed to operate at two different frequency bands, allowing it to support multiple radar frequencies. This improves the flexibility and performance of SAR imaging systems.

2. Dual Polarization

Dual polarization enables the antenna to transmit and receive signals in two orthogonal orientations. This helps in:

- Reducing interference
- Improving signal diversity
- Enhancing overall system reliability

3. Antenna Array Configuration

Instead of using a single antenna element, an array structure is employed. This results in:

- Higher gain
- Improved directivity
- Better radiation characteristics

4. Corporate Feeding Network

A corporate feed network is used to distribute power equally among all antenna elements. This ensures:

- Uniform excitation
- Stable performance
- Better impedance matching

Application Areas

The designed antenna system is highly suitable for:

- Synthetic Aperture Radar (SAR) imaging
- Remote sensing systems
- Satellite communication
- Defense and surveillance applications

Software Used

The antenna design and simulation are carried out using:

CST Microwave Studio, a powerful electromagnetic simulation tool used for analyzing antenna performance parameters such as return loss, gain, and radiation patterns.

LITERATURE REVIEW

In recent years, significant research has been carried out in the field of microstrip antenna arrays, especially for applications in MIMO (Multiple Input Multiple Output) and Synthetic Aperture Radar (SAR) systems. Researchers have focused on improving antenna performance in terms of bandwidth, gain, polarization diversity, and compactness.

Traditional microstrip antennas are widely used due to their simplicity and ease of fabrication. However, they suffer from limitations such as narrow bandwidth and low efficiency, which restrict their use in advanced radar systems. To address these issues, various techniques have been proposed.

One approach involves the use of antenna arrays, where multiple radiating elements are combined to improve gain and directivity. Array configurations such as linear and planar arrays have been studied extensively to achieve better radiation performance.

Another important advancement is the introduction of dual-band antennas, which allow operation at two different frequency ranges. This is particularly useful in SAR systems where multiple frequency bands are required for enhanced imaging and target detection.

In addition, dual-polarized antennas have gained attention due to their ability to transmit and receive signals in two orthogonal polarizations. This improves signal diversity, reduces multipath interference, and enhances overall system reliability.

To support multiple antenna elements in an array, different feeding techniques have been explored. Among them, the corporate feeding network is widely used because it ensures equal power distribution and better impedance matching across all elements.

Recent studies have also focused on compact antenna designs, where the size of the antenna is reduced without significantly affecting performance. This is essential for modern applications where space and weight constraints are critical.

The research paper on which this work is based proposes a compact dual-band dual-polarized microstrip antenna array specifically designed for MIMO-SAR applications. The design integrates multiple advanced features such as:

- Dual-band operation
- Dual polarization
- Array configuration
- Corporate feeding mechanism

These features collectively enhance the antenna performance and make it suitable for high-resolution radar imaging systems.

Summary of Literature

From the reviewed studies, it can be concluded that:

- Microstrip antennas are suitable but require enhancements
- Dual-band and dual-polarization techniques improve performance
- Antenna arrays increase gain and efficiency
- Corporate feeding networks provide better power distribution

The proposed work builds upon these concepts and aims to develop a compact, efficient, and high-performance antenna system for MIMO-SAR applications.

OBJECTIVE OF THE WORK

The primary objective of this project is to design and implement a compact dual-band dual-polarized microstrip antenna array suitable for MIMO-SAR applications.

The specific objectives of the work are as follows:

- To design a microstrip antenna array with a compact structure using suitable substrate materials.
- To achieve dual-band operation, enabling the antenna to function efficiently at two different frequency ranges required for radar applications.
- To incorporate dual polarization, allowing the antenna to transmit and receive signals in two orthogonal orientations for improved signal diversity and reduced interference.
- To develop an efficient corporate feeding network for equal power distribution among all antenna elements.
- To ensure good impedance matching by minimizing return loss (S_{11} parameter) at both operating frequencies.
- To enhance the gain and radiation performance of the antenna using array configuration.

To analyze key antenna parameters such as:

- Return Loss (S-Parameters)
- Gain
- Bandwidth
- Radiation Pattern
- Isolation between ports

To implement and simulate the antenna design using CST Microwave Studio.

To evaluate the suitability of the designed antenna for applications in:

- Synthetic Aperture Radar (SAR)
- Remote sensing
- Satellite communication

METHODOLOGY

The methodology of this project involves a systematic approach to design, simulate, and analyze a dual-band dual-polarized microstrip antenna array. The complete process is carried out using CST Microwave Studio, following step-by-step implementation.

Antenna Design Overview

The antenna is designed as a microstrip patch array, where multiple radiating elements are arranged in a specific configuration to improve gain and directivity.

The design focuses on:

- Compact size
- Dual-band operation
- Dual polarization
- Efficient feeding mechanism

Substrate Selection

The choice of substrate plays a crucial role in antenna performance.

In this design:

- A dielectric substrate (such as FR4 / Rogers RO4350) is used
- Thickness is selected based on design requirements (e.g., 0.508 mm for high-frequency layer)
- Dielectric constant affects:
 - Resonant frequency
 - Bandwidth
 - Efficiency

A proper substrate ensures stable performance and ease of fabrication.

Patch Design

1. The basic radiating element is a rectangular microstrip patch.
2. Steps involved:

- Length (L)
- Width (W)

3. Adjustment of dimensions to achieve resonance at desired frequencies

The patch is designed such that it supports:

- Two resonant modes (dual-band behavior)
- Proper current distribution

Feeding Network Design

A corporate feeding network is used to distribute power equally among all antenna elements.

Key features:

- Uses transmission lines of different impedances
- Includes 50 Ω and 35 Ω microstrip lines
- Ensures equal phase and amplitude

Steps:

- Start from input port (50 Ω)
- Divide power using T-junctions
- Match impedance at each stage

This improves:

- Power distribution
- Impedance matching
- Overall antenna efficiency

Dual-Band Operation

Dual-band behavior is achieved by:

- Modifying patch geometry
- Introducing slots or structural variations

• Benefits:

- Supports multiple radar frequencies
- Improves system flexibility

This allows the antenna to resonate at:

- Lower frequency band
- Higher frequency band

Dual Polarization Technique

Dual polarization is implemented to allow signal transmission in two orthogonal directions.

This is achieved by:

- Proper orientation of feed lines
- Symmetrical antenna structure
- Advantages:
- Reduces interference
- Improves signal quality
- Enhances MIMO performance

Array Configuration

Instead of a single patch, multiple patches are arranged to form an antenna array.

Benefits:

- Increased gain
- Better directivity
- Improved radiation pattern
-

The array is designed with:

- Equal spacing between elements
- Proper alignment for uniform radiation

Simulation in CST

The complete antenna structure is modeled in CST Microwave Studio.

Steps followed:

1. Geometry creation
2. Material assignment
3. Port definition
4. Boundary conditions setup
5. Frequency range selection
6. Simulation run

Parameters analyzed:

1. Return Loss (S11)
2. Gain
3. Radiation Pattern
4. Bandwidth

SIMULATION SETUP (CST MICROWAVE STUDIO)

Simulation Setup

The designed antenna is simulated using CST Microwave Studio, a widely used electromagnetic simulation software for analyzing high-frequency structures such as antennas and RF components.

The simulation process involves creating the antenna geometry, assigning materials, defining ports, and analyzing performance over the desired frequency range.

Software Tool

- **Software Used:** CST Microwave Studio
- **Purpose:** 3D electromagnetic simulation and analysis
- **Application:** Antenna design, S-parameters, radiation characteristics

Geometry Creation

The first step involves modeling the antenna structure in CST:

- Design of rectangular microstrip patches
- Arrangement of patches in an array configuration
- Creation of corporate feeding network using microstrip lines
- Definition of ground plane and substrate layers

All dimensions are defined according to calculated values and optimized for performance.

Material Assignment

Different materials are assigned to each part of the antenna:

- Patch and Feed Lines: Copper (conducting material)
- Substrate: Rogers RO4350 / FR4
- Ground Plane: Perfect Electric Conductor (PEC)

Materials properties such as dielectric constant and thickness are carefully specified.

Port Definition

To excite the antenna:

- A waveguide port or discrete port is assigned at the input feed line
- Input impedance is set to $50\ \Omega$

This allows proper signal excitation and measurement of S-parameters.

Boundary Conditions

Boundary conditions are applied to simulate free-space behavior:

- Open (Add Space) boundary is used
- This ensures accurate radiation simulation

Frequency Range Selection

The simulation is performed over a frequency range that covers both operating bands.

Frequency sweep includes:

- Lower frequency band
- Higher frequency band

This helps in analyzing dual-band performance.

Mesh Generation

1. The structure is divided into small elements (mesh)
2. Finer mesh is used near:
 - Patch edges
 - Feed lines

This improves simulation accuracy.

Parameters Analyzed

After simulation, the following parameters are evaluated:

- Return Loss (S11) → For impedance matching
- Gain → For radiation strength
- Radiation Pattern → For directional behavior
- Bandwidth → Frequency range of operation
- Isolation (S21) → Between antenna ports

RESULTS AND ANALYSIS

After designing and simulating the antenna in CST Microwave Studio, various performance parameters were analyzed to evaluate the effectiveness of the proposed antenna system.

The results confirm that the designed antenna successfully achieves dual-band operation, dual polarization, and improved radiation performance, making it suitable for MIMO-SAR applications.

Return Loss (S-Parameters)

Return loss (S11) is one of the most important parameters used to evaluate antenna performance.

- It indicates how much power is reflected back from the antenna
- A value of $S_{11} < -10$ dB is considered acceptable

Observations:

- The antenna shows two distinct resonant frequencies, confirming dual-band operation
- At both frequencies, the return loss is below -10 dB, indicating good impedance matching
- Minimal reflection ensures efficient power transmission

Conclusion:

- The antenna achieves proper matching at both operating bands, validating the design.

Gain and Radiation Pattern

Gain represents the ability of the antenna to radiate energy in a particular direction.

Observations:

- The antenna array provides higher gain compared to a single patch
- Radiation pattern is directional, which is ideal for radar applications
- Stable radiation is observed at both frequency bands

Radiation Characteristics:

- Main lobe is well-defined
- Side lobes are minimized
- Suitable for SAR imaging, where focused beams are required

Bandwidth Analysis

Bandwidth defines the range of frequencies over which the antenna operates effectively.

Observations:

- The antenna exhibits dual-band bandwidth, covering both frequency ranges
- Each band satisfies the condition where $S_{11} < -10$ dB

Conclusion:

The antenna provides sufficient bandwidth for practical radar and communication applications.

Isolation Performance (MIMO Parameter)

In MIMO systems, isolation between antenna elements is crucial.

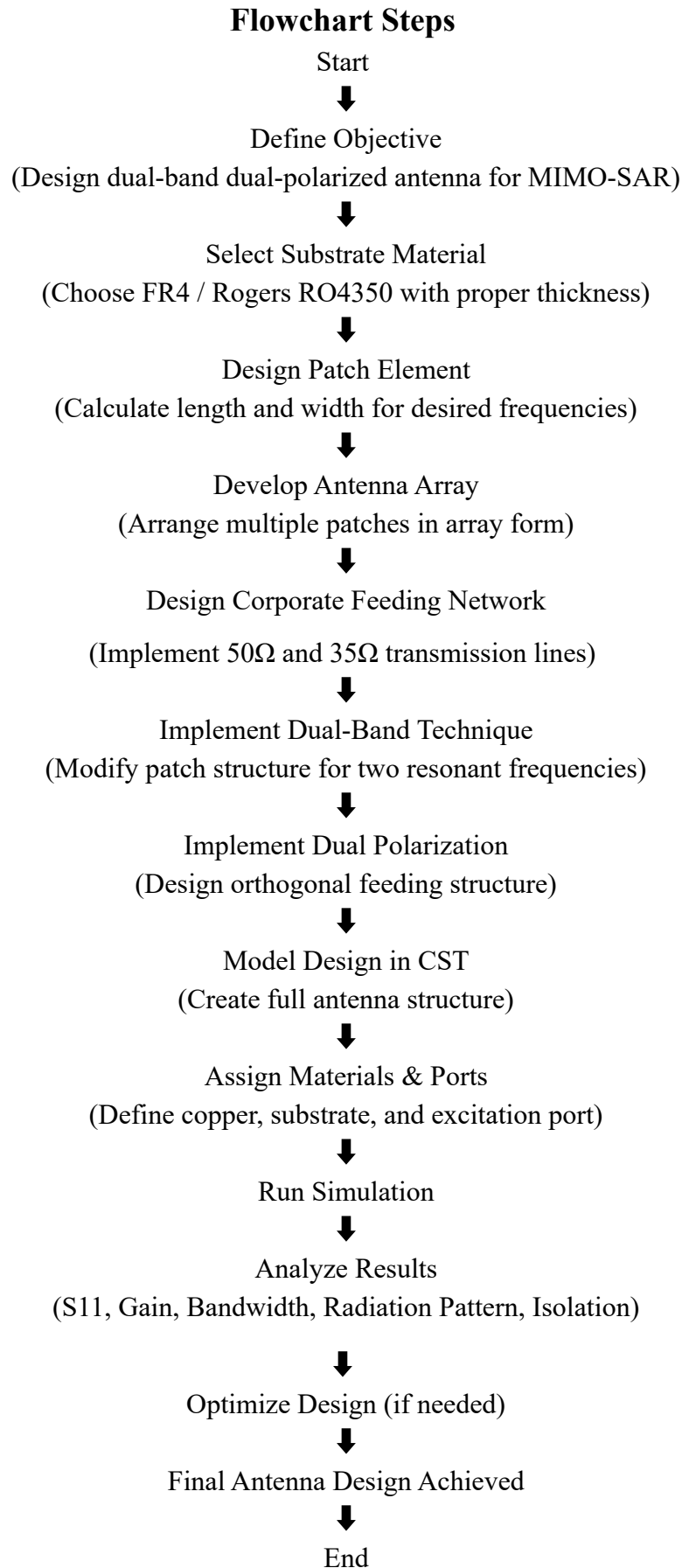
- Measured using S_{21} parameter
- Lower values indicate better isolation
- Observations:
 - The antenna shows good isolation between ports
 - Mutual coupling is minimized
 -

Conclusion:

This ensures proper functioning of MIMO system without interference between channels.

FLOWCHART OF IMPLEMENTATION

The following flowchart represents the step-by-step procedure followed during the design and implementation of the antenna system:



ADVANTAGES AND APPLICATIONS

Advantages of the Proposed Antenna

The designed dual-band dual-polarized microstrip antenna array offers several advantages over conventional antenna systems:

1. Dual-Band Operation

- Supports operation at two different frequency bands
- Enhances flexibility in radar and communication systems

2. Dual Polarization

- Enables transmission and reception in two orthogonal directions
- Reduces signal interference
- Improves signal diversity and reliability

3. Compact Size

- Microstrip technology ensures a low-profile and compact design
- Suitable for systems with space constraints

4. High Gain

- Array configuration significantly improves gain
- Provides better signal strength and coverage

5. Improved Directivity

- Directional radiation pattern
- Ideal for applications requiring focused signal transmission

6. Efficient Power Distribution

- Corporate feeding network ensures equal power to all elements
- Enhances overall antenna performance

Applications

The proposed antenna is highly suitable for various advanced applications:

1. Synthetic Aperture Radar (SAR)

- Used for high-resolution imaging
- Suitable for Earth observation and mapping

2. MIMO Radar Systems

- Supports multiple input and output channels
- Improves detection accuracy and resolution

3. Remote Sensing

- Used in environmental monitoring
- Helps in weather forecasting and terrain analysis

4. Satellite Communication

- Compact and efficient design makes it suitable for space applications

5. Defense and Surveillance

- Used in radar-based detection systems
- Helps in tracking and monitoring targets

CHALLENGES FACED

During the design and implementation of the dual-band dual-polarized microstrip antenna array, several challenges were encountered. These challenges are a natural part of working on advanced antenna systems and helped in gaining practical understanding.

1. Accurate Dimension Calculation

- Determining precise values for patch dimensions (length and width) was challenging
- Small changes in dimensions significantly affected resonant frequency

2. Dual-Band Implementation

- Achieving two resonant frequencies in a single antenna structure required careful design modifications
- Maintaining good performance at both bands simultaneously was difficult

3. Impedance Matching

- Ensuring $S_{11} < -10$ dB at both frequency bands required multiple design iterations
- Proper matching of 50 Ω and 35 Ω transmission lines was complex

4. Corporate Feeding Network Design

- Designing the feed network with equal power distribution was challenging
- Maintaining proper impedance at each junction required careful tuning
-

5. Mutual Coupling (Isolation Issue)

- In antenna arrays, interaction between elements caused unwanted coupling
- Achieving good isolation (low S_{21}) required optimization
-

6. Simulation Complexity in CST

- Setting up correct parameters in CST (ports, boundaries, mesh) required attention
- Simulation time increased due to complex structure

7. Optimization Process

- Multiple iterations were needed to achieve desired results
- Balancing gain, bandwidth, and matching was challenging

8. Understanding Advanced Concepts

- Concepts like dual polarization, MIMO systems, and SAR applications required deep understanding
- Initial learning curve was high

CONCLUSION

In this project, a compact dual-band dual-polarized microstrip antenna array has been successfully designed and implemented for MIMO-SAR applications. The antenna system was developed using microstrip technology and simulated using CST Microwave Studio.

The proposed design effectively achieves dual-band operation, allowing the antenna to function at two different frequency ranges. This enhances the flexibility and usability of the antenna in advanced radar systems. Additionally, the incorporation of dual polarization improves signal diversity and reduces interference, which is essential for efficient MIMO performance.

The use of an antenna array configuration significantly improves the gain and directivity of the system, making it suitable for applications such as SAR imaging and remote sensing. The implementation of a corporate feeding network ensures equal power distribution among all elements, leading to stable and reliable performance.

The simulation results demonstrate that the antenna achieves:

- Good impedance matching ($S_{11} < -10$ dB)
- Adequate bandwidth at both frequency bands
- High gain and directional radiation pattern
- Good isolation between antenna elements
-

Despite several challenges during the design and optimization process, the final antenna structure meets all the required performance criteria.

Final Remark

This project provided valuable hands-on experience in:

- Antenna design and analysis
- Use of CST simulation software
- Understanding of MIMO and SAR systems

The developed antenna system proves to be a compact, efficient, and high-performance solution for modern radar and communication applications.

FUTURE SCOPE

The proposed dual-band dual-polarized microstrip antenna array demonstrates promising performance for MIMO-SAR applications. However, there are several areas where further improvements and advancements can be made.

1. Fabrication and Testing

- The antenna can be physically fabricated using PCB technology
- Real-time measurements can be performed to validate simulation results

2. Bandwidth Enhancement

- Techniques such as slotting, stacking, or use of advanced substrates can be applied
- This can further improve bandwidth and efficiency

3. Gain Improvement

- Increasing the number of array elements can enhance gain
- Optimization of array spacing can improve radiation performance

4. Miniaturization

- Advanced design techniques can be used to further reduce antenna size
- Useful for compact and portable devices

5. Advanced MIMO Systems

- The design can be extended for higher-order MIMO systems
- Can support more input-output channels for better performance

6. Integration with Modern Technologies

- Can be integrated with 5G and future wireless communication systems
- Suitable for next-generation radar and sensing technologies

7. Use of AI-Based Optimization

- Machine learning techniques can be used to optimize antenna parameters
- Reduces design time and improves performance accuracy

REFERENCES

The following references were used during the study, design, and implementation of the antenna system:

Research Papers:

1.A Compact Dual-Band Dual-Polarized Microstrip Antenna Array for MIMO-SAR Applications

(Primary research paper used for this project)

Books:

2.C. A. Balanis,

“Antenna Theory: Analysis and Design”,
Wiley Publications

3. Ramesh Garg, Prakash Bhartia, Inder Bahl, and Apisak Ittipiboon,

“Microstrip Antenna Design Handbook”,
Artech House

Software Tool:

4.CST Microwave Studio

Used for antenna design, simulation, and analysis